MACROECONOMIC & PHILOSOPHICAL ANALYSIS REPORT

Subject: Compatibility of the RCED v5 model with planetary limits, local resources, and space exploration

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Status: Operational Synthesis Document (Parts I & II)

Foreword

This report aims to synthesize the physical, ecological, and technological implications of the Automation Gains Distribution model (RCED v5). It analyzes how the model pivots in response to the dual constraints of Earth's finitude and the call of the spatial horizon.

1. Model Synthesis (Part I: The Foundation)

The RCED v5 model relies on a strict distribution engine for productivity gains derived from automation ($r \approx 5.3\%$ annually on the reference trajectory), governed by the immutable golden rule:

$$\alpha + \beta + \gamma + \delta = 1$$

Where **α** (liberated time), **β** (Sovereign Wealth Fund), **γ** (sectoral transition), and **δ** (Decarbonization Fund) absorb the entirety of the created value without relying on inflation or the infinite expansion of the money supply.

2. Compatibility with Planetary and Local Limits

The RCED v5 model rejects the dogma of "green growth" through hypothetical decoupling. It integrates the systemic finitude of the Earth as an absolute constraint variable.

A. Planned Reduction of the Material Footprint

Unlike classical capitalist indicators, the model seeks to **decarbonize by actively reducing final energy consumption** (targeting a 22.6% decrease by Year 20).

- **Relocation of flows:** The reduction in working hours (α = 45%) and the usage fee short-circuit hyper-extended global supply chains in favor of bioregional short circuits.
- **Mass efficiency:** The deployment of the *Citizen Transition Service* (Layer 13) allows for the thermal insulation of the national building stock in record time, eliminating the need for energy overproduction.

B. Independence from Critical Minerals (Layer 13)

The major trap of the global technological transition lies in its dependence on rare metals (Lithium, Cobalt, Neodymium). RCED v5 bypasses this limit through a high-engineering low-tech approach:

- **Stationary thermal storage:** Heating (45% of the demand in temperate countries) is shifted to high-temperature **sand batteries**.
- **Geopolitical sobriety:** Sand is a local, abundant, and infinitely recyclable resource, avoiding colonialized extractivism in the Global South.

C. Fractal Network Optimization (Layer 14)

By applying the principle of **energy subsidiarity** via nested cells (building → neighborhood → region), peer-to-peer energy sharing smooths out intermittency.

> ***Key Indicator:*** *This local mesh limits the required infrastructure overbuilding (overbuild) to a factor of **1.25x**, compared to over 2x for a traditional centralized grid. Thousands of tons of copper and silicon are thus conserved.*

D. The End of Hoarding: Usage Property (Patch v4.4.3)

The legal framework eliminates speculation on physical resources. A registry of actual use replaces speculative property titles. If a material resource (land, infrastructure, energy) is not utilized for the common good or active use, the right of retention lapses, preventing waste and over-extraction.

3. The Space Trial: Conquest vs. Exploration

Extending the RCED v5 model to the spatial question requires a major conceptual distinction between the current extractivist model and the Commons model.

A. Absolute Incompatibility with Commercial *New Space*

The model structurally blocks the trajectory of a privatized space conquest akin to Elon Musk's vision:

- **Disappearance of mega-fortunes:** The distribution model reinjects technological gains back into society. The capital accumulation required for an individual to afford a private space program becomes impossible.
- **Non-existence of extraterrestrial property rights:** Patch v4.4.3 forbids claiming private ownership of an asteroid or planetary soil. For-profit space mining is constitutionlessly illegal.

B. The Democratic Filter of Payload Allocation

In Part I, space is no longer a financial investment, but a **carbon and material allocation trade-off** overseen by the Citizen Assembly. Every rocket launch must be validated against the maintenance needs of Earth's biosphere. Consequently, space undergoes a de facto moratorium on commercial and tourist flights.

C. The "Star Trek" Horizon (Part II: Federation Earth)

In the long term, once terrestrial civilization is stabilized (net-zero emissions, Gini coefficient below 0.200, operational post-scarcity), space exploration is reborn in a purely scientific form.

- **Celestial Bodies as Universal Commons:** Space is managed as an extension of Layer 11 (Global Commons). Any resource harvested from space is captured by Federation Earth to fund the global **Commons Dividend**.
- **Pure Curiosity as a Driver:** Liberated from the necessity of profit, humanity uses its automation surpluses to explore the universe—not to flee a ruined Earth, but out of a pure ambition for knowledge.

Conclusion

The RCED v5 model demonstrates that in the short term, it safeguards planetary limits by replacing the incentive for growth with the incentive for liberated time. In the long term, it prevents the transplantation of predatory capitalism into space and prepares humanity for ethical exploration under the banner of the Universal Commons.